Structuring Human Objectives: A Survey of Rubrics for Evaluation, Alignment, and Agentic AI

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https://github.com/FreedomIntelligence/Awesome-Rubrics

Abstract

Evaluation and feedback design has become a central bottleneck in post-training foundation models for open-ended, professional, multimodal, and agentic tasks. Conventional signals, such as reference answers, scalar ratings, pairwise preferences, and exact verifiers, are often insufficient for multi-dimensional, interpretable, and context-dependent assessment. To address this gap, rubric-based supervision decomposes complex human objectives into explicit and reusable criteria. However, existing studies remain fragmented across rubric construction, evaluation, reward design, and agent assessment, while a unified understanding of rubrics is still lacking. This paper provides a comprehensive survey of rubric-centered research for LLM evaluation and post-training. We present a unified view of rubrics as a structured supervision interface between human judgment and model optimization, and propose a systematic taxonomy covering data construction, training integration, inference enhancement, risks, evaluation mechanisms, and applications. Furthermore, we discuss how rubrics support scalable evaluation, reward modeling, judge alignment, and highlight key challenges and future directions.

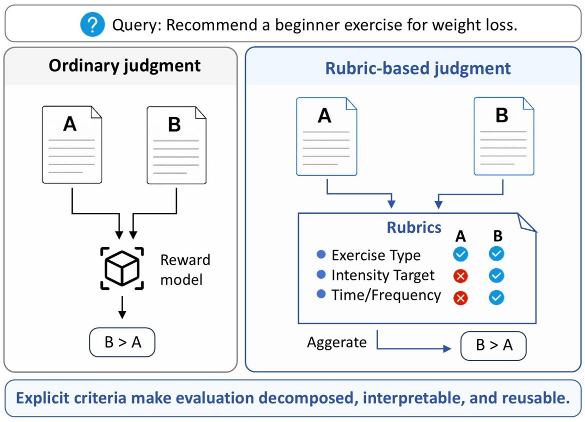


Figure 1: Comparison between ordinary judgment and rubric-based judgment. Rubrics decompose expert expectations into explicit criteria, enabling more interpretable evaluation and reusable supervision signals.

1 Introduction

Recent advances in large language models (LLMs) are reshaping AI systems from static conversational agents toward increasingly complex reasoning and agentic systems (Gao et al., 2026b). Early chatbot models primarily focused on response fluency and relevance in single-turn interactions. More recent reasoning-oriented models emphasize logical consistency and correctness on complex reasoning tasks. However, as LLMs evolve into autonomous agents capable of long-horizon planning, tool use, and real-world interaction, the optimization target itself becomes fundamentally more complex (Yang et al., 2026c; Zhang et al., 2026f). Modern AI systems are no longer evaluated solely by answer correctness or human preference. Instead, they are expected to satisfy *diverse and often competing objectives*, including safety, efficiency, planning quality, policy compliance, robustness, interaction quality, and cost constraints. Consequently, the central challenge of AI systems is gradually shifting from scaling model capacity to scaling supervision and evaluation (Gunjal et al., 2026).

This transition reveals the limits of traditional supervision. While labels, references, scalar ratings, and pairwise preferences work well for clearly defined objectives, they become insufficient for open-ended generation, long-horizon reasoning, multi-step interaction, and agentic tasks (Shao et al., 2025; Li et al., 2026d). In these settings, a single holistic signal cannot explain why an output succeeds or fails, which constraints are met or violated, or where errors arise. Compressing complex human expectations into one score or preference therefore causes information loss, weak interpretability, and unstable optimization, especially when real-world objectives are structured and compositional (Zhang et al., 2026g).

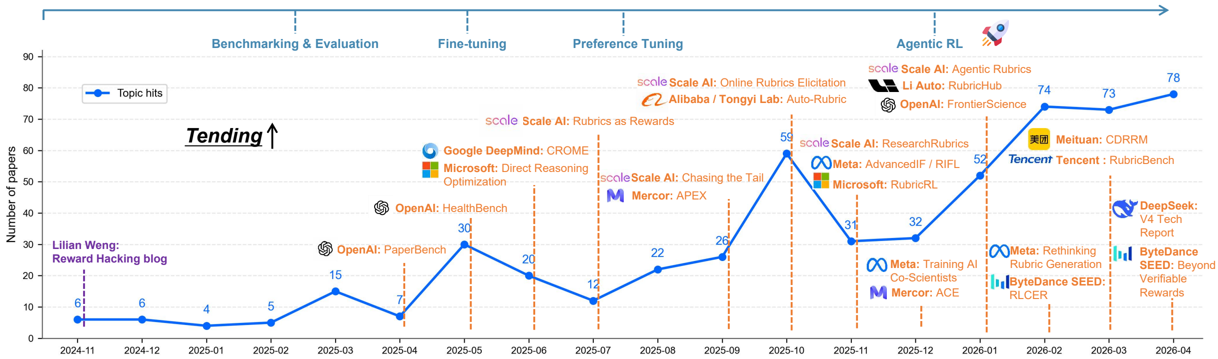


Figure 2: Recent trend of rubric-based reinforcement learning. The increasing number of studies, especially from industrial laboratories, reflects the growing role of rubrics in scalable evaluation, reward design, and model alignment.

Against this backdrop, rubrics have emerged as a promising paradigm for structured supervision and evaluation (Lyu et al., 2026a). Rather than treating quality as monolithic, rubrics decompose complex objectives into explicit, inspectable, and composable criteria (Rezaei et al., 2025). They offer an **effective trade-off** between expert knowledge and model-based evaluation by translating human expertise into structured criteria that can be repeatedly applied by scalable evaluators. In this way, rubrics transform implicit human expectations into machine-operational signals, supporting evaluation, reward modeling, preference construction, reinforcement learning, data annotation, and error diagnosis (Zhang et al., 2026e).

The growing importance of rubrics reflects a broader shift in modern AI development: **evaluation is becoming core infrastructure** for building and aligning AI systems. Evaluation no longer merely measures model outputs after deployment. It increasingly defines optimization targets, failure penalties, and training rewards (Fan et al., 2025). Yet high-quality evaluation often relies on expert experience, including domain norms, safety constraints, task priorities, and practical failure modes (Zhang et al., 2026b). The key challenge is therefore to **externalize and operationalize expert judgment** into criteria that can be repeatedly applied by humans, models, or hybrid evaluators. As illustrated in Figure 1, holistic judgment provides only a coarse preference comparison, such as whether response B is better than response A. Rubric-based judgment instead decomposes this preference into explicit criteria, making the evaluation more interpretable, reusable, and suitable for reward construction.

This shift is reflected in recent trends. As shown in Figure 2, rubric-related reinforcement learning has grown rapidly, with increasing participation from major **industrial laboratories**, indicating that rubrics are emerging as practical mechanisms for scalable reward design and model alignment.

Despite growing interest, rubric research remains fragmented across *sources*, *automatic generation*, *evaluation*, *multi-objective rewards*, *process supervision*, and *agent assessment*. This leaves several foundational questions underexplored:

❶ What constitutes a high-quality rubric?

❷ How should rubrics balance granularity, coverage, and executability?

❸ How can they be automatically generated while maintaining reliability and interpretability?

❹ How do they influence optimization dynamics, model behavior, and reward hacking?

❺ More broadly, when and why do rubrics provide effective structured supervision?

To answer these questions, this survey presents a unified view of rubrics as **a structured supervision paradigm for complex AI systems**. Rather than serving solely as evaluation criteria or annotation templates, rubrics translate human objectives into interpretable, compositional, and optimizable signals. As shown in Fig. 3, we systematically review the literature across six dimensions: **data** construction, **training** integration, **inference** enhancement, potential **risk**, **evaluation mechanisms**, and **applications**.

2 Preliminaries

2.1 Basic Concepts

Rubrics were originally developed in educational assessment as explicit scoring guides for improving evaluation transparency and inter-rater consistency (Panadero and Jonsson, 2013). In LLM systems, rubrics have evolved into a supervision in-

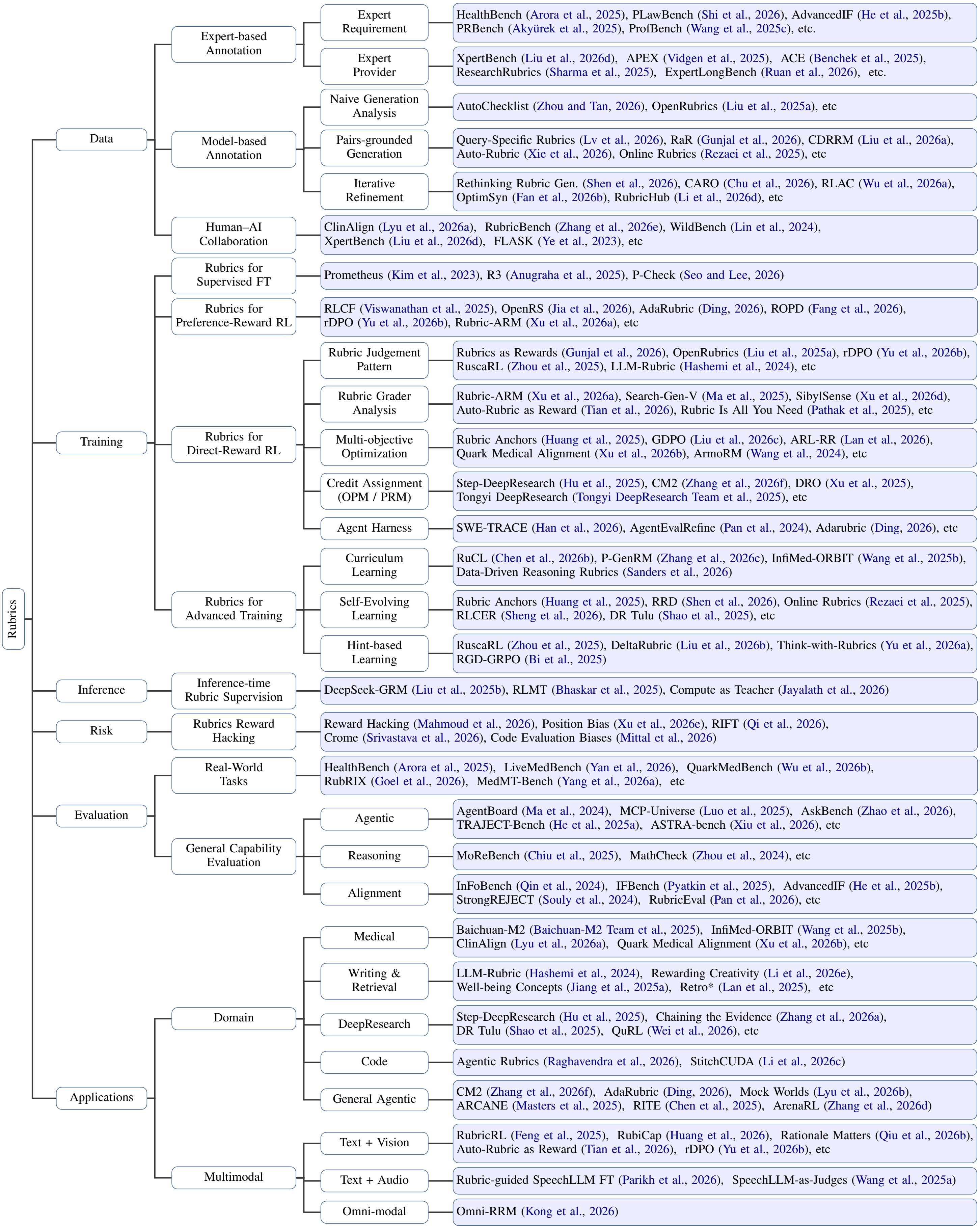


Figure 3: Taxonomy of rubrics in the era of large models.

Principle	Core Character	✗ Bad Rubric	✓ Good Rubric
Atomicity	**A single criterion** should evaluate only **one requirement**.	"Does the response ask about *symptoms, causes, and treatment*?"	"Does the response ask about the duration of the **skin symptom**?"
Context Specificity	Criteria should be **self-contained** and **avoid unclear references**.	"For a skin-rash query, does the response ask about *disease details*?"	"For a skin-rash query, does the response ask about **rash location or itchiness**?"
Checkability	Criteria should be directly **judgeable** from **observable evidence**.	"Is the response *appropriate, detailed, and professional*?"	"Does the response explicitly ask whether the **rash is itchy**?"
Completeness	Rubrics should **cover key constraints** and **exclude irrelevant ones**.	"For a phone-repair query, does the response state the *7-day return policy*?"	"For a phone-repair query, does the response provide the **repair contact number**?"
Correctness	Criteria should **avoid factual errors** and **invalid reasoning**.	"Does the answer state that BERT *is a decoder-only language model*?"	"Does the answer state that BERT **is an encoder-based language model**?"

Table 1: Examples of good and bad rubrics under different design principles.

terface that makes evaluation criteria explicit: they define which properties of a response should be examined, how each property should be judged, and how criterion-level judgments can be used for evaluation or post-training (Zhang et al., 2026b). The interface is particularly valuable for open-ended tasks, where response quality depends not only on answer correctness, but also on factuality, coverage, reasoning, safety, style, tool use, and domain-specific constraints (Gunjal et al., 2026).

Core Components Rubric supervision can operate at different granularities, including final answers, reasoning steps, tool actions, and full trajectories (Gunjal et al., 2026). A typical pipeline consists of five components: (1) ***Task preparation***, which defines prompts, references, user intent, or domain constraints; (2) ***Rubric construction***, which derives explicit criteria from expert knowledge, references, preference data, retrieved evidence, or observed failures (Xie et al., 2026); (3) ***Response rollout***, which samples answers, reasoning traces, or agent trajectories; (4) ***Criterion-level judging***, which produces criterion-wise assessments using humans, LLM judges, reward models, or verifiers (Kim et al., 2023); and (5) ***Aggregation and use***, which combines these assessments for evaluation, error analysis, reward modeling, and optimization (Zhang et al., 2026b).

Formal Definition The above pipeline can be formalized as follows. Let x be a task prompt and $y \sim \pi(\cdot \mid x)$ a model response or trajectory. A prompt-conditioned rubric is defined as

$$\mathcal{R}(x) = \{(c_j, a_j, w_j, m_j)\}_{j=1}^{K}, \tag{1}$$

where c_j is a natural-language criterion, a_j denotes its judgment space, such as binary labels, ordinal levels, or numeric scores, w_j is an optional weight, and m_j records metadata such as dimension, evidence source, reward granularity, or pitfall type. Given an evaluator V, each criterion receives a judgment

$$z_j = V(x, y, c_j) \in a_j. \tag{2}$$

For non-numeric judgments, z_j can be mapped to a numeric score before aggregation. The final rubric score is then computed by an aggregation function:

$$S_{\mathcal{R}}(x, y) = A(\{(z_j, w_j, m_j)\}_{j=1}^{K}). \tag{3}$$

Here, A can be instantiated as a weighted average, rule-based gate, learned aggregator, or implicit judge aggregation. When used for policy optimization, the rubric score can serve as the reward signal, i.e., $r(x, y) = S_{\mathcal{R}}(x, y)$.

2.2 Principles of Good Rubrics

To better leverage rubrics as effective supervision signals, it is necessary to clarify what properties reliable rubric data should satisfy. Regardless of whether rubrics are written by experts or generated by models, high-quality rubric criteria should be clear, verifiable, and aligned with the target task (Zhang et al., 2026e). As summarized in Table 1, we highlight five key principles:

(1) ***Atomicity*** requires each criterion to assess only one requirement, so that it can be judged independently (Chen et al., 2026a); (2) ***Context specificity*** requires criteria to provide sufficient task context and avoid vague references such as "disease details" or "the issue" (Arora et al., 2025); (3) ***Checkability*** requires criteria to be directly judgeable from observable evidence rather than broad subjective standards such as "appropriate" or "professional" (He et al., 2025b); (4) ***Completeness*** requires the rubric set to cover key task constraints while excluding irrelevant requirements (Akyürek

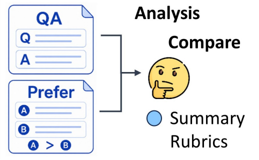
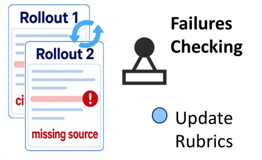
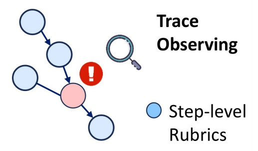
	Expert-defined	Reference-grounded	Failure-driven	Process-aware
Illustration				
Desc.	Rubrics are annotated by **domain experts** to encode professional standards.	Rubrics are distilled from **references** to capture expected content.	Rubrics are updated from **failures** by rewriting errors into criteria.	Rubrics are derived from **process traces** to specify step-level requirements.
Works	HealthBench (Arora et al., 2025), APEX (Vidgen et al., 2025).	OpenRubrics (Liu et al., 2025a), ClinAlign (Lyu et al., 2026a).	RLAC (Wu et al., 2026a), Online Rubrics (Rezaei et al., 2025).	Step-DeepResearch (Hu et al., 2025); SWE-TRACE (Han et al., 2026).

Table 2: Illustration of representative sources for rubric data construction.

et al., 2025); and (5) ***Correctness*** requires criteria to avoid factual errors and invalid reasoning, since flawed criteria can systematically reward incorrect responses (Shi et al., 2026). Together, these principles make rubrics more reliable and auditable as supervision signals.

❶ A high-quality rubric should define atomic, context-specific, verifiable, complete, and correct criteria, thereby making human objectives explicit, judgeable, and auditable.

3 Rubric Data Construction

Rubric data construction transforms diverse supervision evidence into explicit and judgeable criteria. This section reviews major rubric sources, compares expert-based, model-based, and human-AI construction paradigms, and discusses how rubric quality is evaluated through alignment and downstream judgment utility.

3.1 Sources for Rubric Data

Before rubrics can be constructed, a central question is where their criteria should come from. Although rubric construction methods vary across tasks and domains, most approaches rely on different forms of supervision evidence to define what should be evaluated. These evidence sources determine the initial content of rubric criteria, while the subsequent construction process further transforms them into reliable rubric data through decomposition, filtering, grounding, and calibration.

Table 2 summarizes four representative evidence sources for rubric data construction. (1) ***Expert-derived*** rubrics use domain specialists to formulate criteria from professional standards, task-specific expertise, and human judgment (Vidgen et al., 2025). This source is especially valuable when correctness depends on specialized norms or domain knowledge. (2) ***Reference-grounded*** rubrics extract criteria from reference answers, gold-standard outputs, or preference annotations, allowing criteria to capture expected content, reasoning processes, and quality dimensions (Lyu et al., 2026a). (3) ***Failure-driven*** rubrics convert observed model errors into explicit criteria, so that recurring mistakes, omissions, or unsafe behaviors can be penalized in future evaluation or optimization (Wu et al., 2026a). (4) ***Process-aware*** rubrics use intermediate reasoning steps, tool-use trajectories, or agent execution traces to define step-level requirements and procedural constraints, thereby extending rubric supervision beyond final answers (Han et al., 2026).

These four evidence sources provide different starting points for rubric data construction. ***Expert-derived*** rubrics emphasize reliability and domain validity. ***Reference-grounded*** rubrics improve alignment with expected answers or human preferences. ***Failure-driven*** rubrics strengthen coverage of known error modes. ***Process-aware*** rubrics support fine-grained supervision over reasoning processes, tool use, and agent behavior.

However, evidence sources alone do not guarantee high-quality rubrics. Raw criteria may still be too coarse, redundant, incomplete, vague, or difficult to judge. Therefore, the following subsections focus on how different construction paradigms transform these sources into usable rubric data through decomposition, filtering, grounding, and calibration.

3.2 Expert-based Annotation

Expert-based annotation prioritizes reliability by grounding rubric criteria in professional norms, domain knowledge, and human judgment. Here, experts balance fine-grained diagnosis, requirement

Type	*Efficiency*	*Scalability*	*Expertise*	*Potential Risk*	*Representative Papers*
Expert-based	✗ Low	✗ Low	✓ High	▲ Annotation bottleneck	HealthBench (Arora et al., 2025), PLAWBench (Shi et al., 2026)
Model-based	✓ High	✓ High	✗ Low	▲ Rubric bias	CDRRM (Liu et al., 2026a), RubricHub (Li et al., 2026d)
Human-AI Hybrid	△ Balanced	△ Balanced	△ Balanced	▲ Calibration burden	ClinAlign (Lyu et al., 2026a), RubricBench (Zhang et al., 2026e)

Table 3: Comparison of three rubric construction paradigms in terms of efficiency, scalability, expertise, potential risk, and representative papers.

coverage, and executable judgment by deciding what domain requirements to include, how detailed criteria should be, and whether standards can be applied consistently. Its main strength is coverage, since experts identify valid answer components, serious omissions, and domain-specific trade-offs. Expert-written rubrics also serve as calibration anchors for generated rubrics and LLM judges, but remain slow, costly, and difficult to scale. We discuss two aspects: (1) ***Expert requirements*** and (2) ***Expert providers***.

Expert Requirements Experts translate domain standards into rubric criteria by specifying essential requirements, critical omissions, practical constraints, and task-specific failure modes (He et al., 2025b; Wang et al., 2025c). These requirements prevent rubrics from being too generic or incomplete by identifying core checks and serious failures. Medical rubrics cover safety, accuracy, communication, and context awareness (Arora et al., 2025), while legal rubrics capture statutory grounding, factual extraction, and reasoning structure (Shi et al., 2026). Long-form and research rubrics further specify factual grounding, synthesis quality, citation use, and reasoning soundness (Akyürek et al., 2025; Chen et al., 2026a). Thus, they calibrate generated rubrics and LLM judges (Arora et al., 2025; He et al., 2025b).

Expert Provider Provider choice affects domain validity, criterion detail, and scoring consistency. In professional domains, rubric writers are often credentialed practitioners or experienced providers who encode task-specific standards, workplace constraints, and domain conventions (Liu et al., 2026d; Vidgen et al., 2025). Yet credentials alone are insufficient: broader expert tasks require annotators who can design realistic prompts, anticipate failure modes, and write stable criteria (Ruan et al., 2026; Sharma et al., 2025). Therefore, expert-rubric datasets should screen annotators, separate drafting from review, and use feedback or arbitration for ambiguous cases (Sharma et al., 2025; Zhang et al., 2026e). Related evaluation pipelines similarly rely on domain providers to validate criteria (Benchek et al., 2025).

❷ Expert annotation balances rubric design through **expert judgment** over required content, decomposition depth, and criterion applicability.

3.3 Model-based Annotation

Model-based annotation improves scalability by generating rubrics quickly and cheaply. Here, automatic generation and refinement approximate the balance among detailed decomposition, requirement coverage, and judgeability: LLMs break broad quality dimensions into instance-specific criteria, while filtering, grounding, and reweighting reduce redundancy, recover missing constraints, and improve applicability. However, generated rubrics may hallucinate requirements, miss constraints, drift from human standards, or become redundant. The key challenge is to preserve reliability and interpretability. Existing work mainly follows three strategies: (1) ***Naive generation***, (2) ***Preference-assisted generation***, and (3) ***Iterative refinement***.

Naive Generation. The simplest approach prompts an LLM to generate rubrics or checklist items from the query (Zhou and Tan, 2026; Dhole and Agichtein, 2026). It is useful for bootstrapping rubric data and structured supervision, especially when no human-written rubric is available. However, because the model must infer task requirements directly from the prompt, generated criteria may be generic, incomplete, or spurious. This can leave implicit requirements uncovered, introduce irrelevant checks, or produce vague criteria that judges cannot apply consistently. When used as rewards, such errors can become optimization targets, making naive generation a scalable baseline rather than a reliable endpoint (Liu et al., 2025a; Gao et al., 2026a).

Preference-assisted Generation. This strategy compares a preferred response y^+ with a rejected response y^- and infers criteria that explain their quality difference (Lv et al., 2026; Liu et al., 2026a). By grounding criteria in observable preference differences, it makes rubric items more concrete and discriminative (Xie et al., 2026; Wang and Xiong, 2025). Such criteria expose failure modes that actually affect human judgments, and can be checked by whether rubric-based judging recovers the original preference label (Rezaei et al., 2025; Gunjal et al., 2026). However, pair-derived rubrics may remain local: they can explain one comparison but fail to capture requirements that only appear in broader future responses (Liu et al., 2026a).

Iterative Refinement. Refinement treats generated rubrics as drafts rather than final standards, checking whether they are complete, independent, verifiable, and useful for judging. Decomposition turns broad criteria into more atomic checks, filtering removes redundant or irrelevant items while preserving key constraints, and reweighting shifts rubrics toward harder or more discriminative dimensions (Shen et al., 2026; Chu et al., 2026). Other refinement pipelines similarly use quality-control loops to improve criterion stability and downstream usefulness (Yang et al., 2026c; Fan et al., 2026b; Li et al., 2026d). By adding this quality-control loop, refinement turns scalable but noisy rubric generation into more stable supervision signals (Shen et al., 2026; Wu et al., 2026a).

❷ Model-based annotation uses **automatic generation and refinement** to decompose, filter, ground, and reweight criteria for judgeability.

3.4 Human-AI Collaboration

Human-AI hybrid annotation combines model scalability with human calibration. In this paradigm, models provide scalable and fine-grained rubric drafts, while humans or expert standards review, correct, ground, and calibrate them to preserve important requirements and judgeability. Unlike purely model-based refinement, hybrid annotation grounds generated criteria in external standards, such as human reviews, expert responses, benchmark rubrics, or clinician preference data (Lyu et al., 2026a; Zhang et al., 2026e). Humans review candidates, remove irrelevant items, calibrate scoring rules, or provide expert demonstrations for criterion distillation. This role is necessary because generated rubrics may be subjective, incomplete, redundant, or too coarse (Zhang et al., 2026e; Liu et al., 2026d). Thus, hybrid pipelines combine model speed with human correction of drift, hallucination, and reward mismatch (Lyu et al., 2026a; Zhang et al., 2026e).

❸ Human-AI collaboration improves reliable, interpretable rubric generation by using **human or expert standards** to review, ground, and calibrate drafts.

3.5 Evaluation on Rubric Quality

Generated rubrics are often evaluated in tasks where outputs are open-ended and cannot be judged by exact-match answers. Typical settings include open-ended generation, instruction-following, preference judgment, and domain-specific assessment tasks. In these scenarios, reference answers are often insufficient, and rubrics are needed to capture multiple quality dimensions, such as completeness, factuality, safety, reasoning quality, and adherence to task constraints.

The goal is to evaluate two complementary properties of rubrics. (1) ***Rubric-level alignment*** measures whether model-generated rubrics are consistent with human-written or expert-validated rubrics, using metrics such as rubric recall, hallucination rate, and structural F1. (2) ***Judgment-level utility*** evaluates whether rubrics are useful as inputs to LLM evaluators by testing whether rubric-guided judgments recover human preferences, often measured by preference accuracy. RubricBench provides a representative example of this evaluation setting (Zhang et al., 2026e). Detailed benchmark designs and metrics are discussed in Section 7.2.

4 Rubric-guided Model Training

Rubrics as training-time supervision. Rubric-guided training examines how structured supervision reshapes post-training when rubrics move from evaluation templates to optimization signals. As illustrated in Figure 4, rubrics enter model training through supervised fine-tuning, preference-reward RL, direct-reward RL, and advanced training. Across these stages, rubrics affect not only what is rewarded, but also how supervision is decomposed, assigned, aggregated, calibrated, and

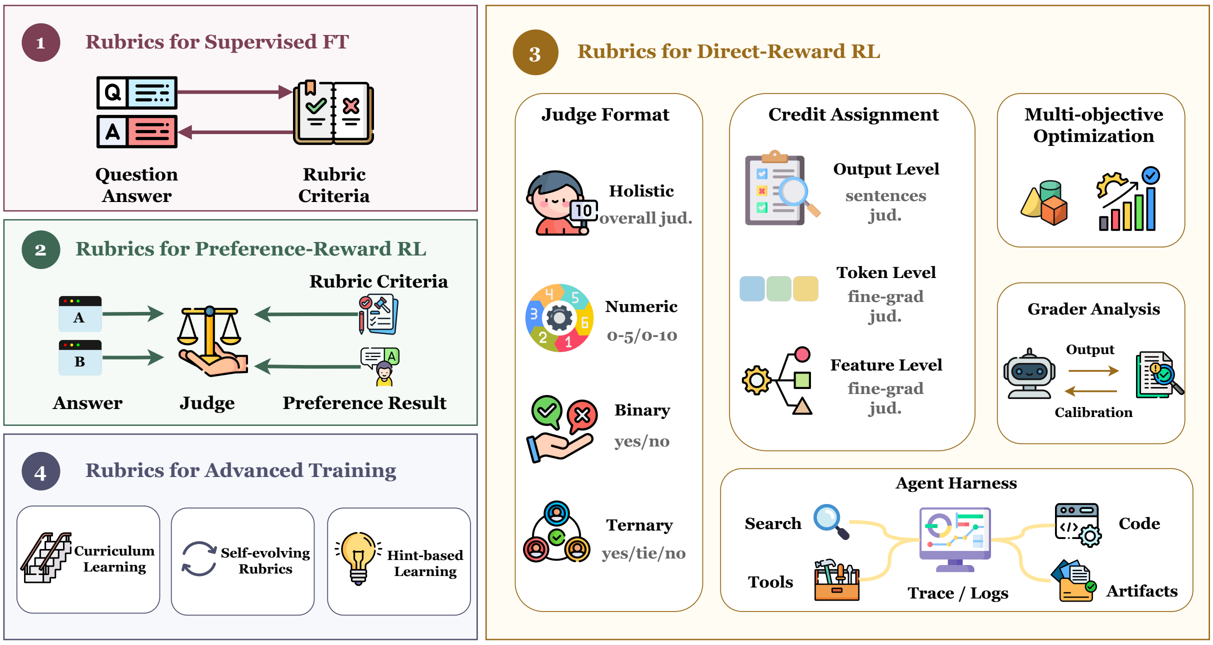


Figure 4: Illustration of common strategies for rubric-based optimization. More details of the related methods please refer to Table 6.

adapted during learning.

4.1 Rubrics for Supervised Fine-Tuning

Rubric-guided supervised fine-tuning (SFT) builds the model-side interface for using rubrics before criteria are converted into optimization signals. Rather than directly optimizing a response policy, it trains models, typically evaluators, reward models, or feedback generators, to produce criterion-conditioned scores, feedback, or rationales. This makes the basis of judgment explicit and enables evaluator-style models with customized scoring rubrics (Kim et al., 2023). The same interface also supports reward modeling and personalization by recasting heterogeneous supervision signals or user histories as rubric-grounded reasoning (Anugraha et al., 2025; Seo and Lee, 2026). Thus, SFT mainly makes rubric criteria executable by models, while stronger behavioral optimization is left to later preference-learning or reinforcement-learning stages.

4.2 Rubrics for Preference-Reward RL

Building on SFT, preference-reward RL is the first stage where rubric supervision begins to shape policy optimization through preference data. It decomposes pairwise chosen–rejected labels into criterion-level judgments for scoring, filtering, or reweighting training pairs (Viswanathan et al., 2025). **Preference decomposition** improves DPO-style training by selecting more informative comparisons and revealing dimension-specific failures hidden by holistic labels (Ding, 2026). **Preference construction** further extends rubric signals to multimodal preference data (Yu et al., 2026b) and on-policy distillation, where teacher–student gaps are converted into prompt-specific rubric rewards (Fang et al., 2026). Together, these methods connect preference data with optimization by making reward signals more interpretable and decomposable, although their effectiveness still depends on the quality of criteria, labels, and LLM judges.

4.3 Rubrics for Direct-Reward RL

Direct-reward RL makes the optimization role of rubrics most explicit: rubric criteria are no longer only reported as evaluation results, but are operationalized as reward signals. A rubric-based RL pipeline must decide how rubric satisfaction is scored, where rewards are assigned, how multiple criteria are aggregated, which grader applies the rubric, and how process traces are captured for agentic tasks. In this setting, judge format defines the feedback interface, credit assignment specifies which behaviors are reinforced, aggregation determines objective trade-offs, grader analysis shapes the reward landscape, and agent harnesses expose process-level supervision.

> ❹ Rubrics influence optimization by shaping **reward formulation**, **grader choice**, objective aggregation, and credit-assignment granularity.

Judge Format. A key design choice is how rubric satisfaction is exposed as reward feedback. Existing formats range from global scores to criterion-level judgments. *Holistic scoring* gives the grader all rubrics and asks for one overall score, which is easy to integrate into RL but leaves criterion weights, partial credit, and violation severity implicit (Gunjal et al., 2026; Liu et al., 2025a). *Numeric scoring* assigns scalar or ordinal criterion-level scores, preserving graded quality differences such as coherence or persuasiveness, but requiring scale calibration across examples (Wu et al., 2025; Hashemi et al., 2024). *Binary judgment* turns each criterion into a yes/no check, improving verifiability and auditability at the cost of partial credit (Rezaei et al., 2025; Arora et al., 2025; Zhou et al., 2025). *Ternary judgment* adds an intermediate state, making it better suited to long-form, search-augmented, or multimodal responses that may be only partially correct or grounded (Shao et al., 2025; Ma et al., 2025). Overall, the field is moving from convenient global scores toward criterion-level, auditable rewards, a shift that becomes especially important once rubric scores are optimized rather than merely reported.

Credit Assignment. Credit assignment determines where rubric rewards are attached during optimization. At the coarsest level, outcome-level rewards assign one score to a complete response or report, which is simple and cheap but sparse: the same scalar must cover evidence use, reasoning detours, synthesis quality, and late-stage hallucinations. Finer-grained methods densify supervision by assigning credit to smaller units. Token-level approaches derive local rewards under rubric-gated constraints, while feature-level approaches extract reward signals from the judge's internal computation or model representations (Xu et al., 2025; Zhang et al., 2026g; Prasad et al., 2026). The central trade-off is density versus faithfulness: denser rewards can improve sample efficiency, but they also introduce finer-grained proxies that must remain aligned with the original criterion-level intent.

Multi-objective Optimization. Rubric-based direct-reward RL is inherently multi-objective, since criteria such as factuality, safety, completeness, concision, evidence use, and task-specific constraints are not fully substitutable. A naive weighted sum may therefore reward superficial gains while masking violations of critical requirements. This motivates anchor-, penalty-, and constraint-based designs that preserve non-negotiable requirements instead of averaging them away (Huang et al., 2025; Xu et al., 2026b).

Recent methods preserve reward structure before scalarization in different ways. *Group-wise optimization* separates reward groups before advantage computation, preventing one dimension from erasing another's learning signal (Liu et al., 2026c). *Context-dependent scalarization* treats rubric dimensions as interpretable objectives whose weights vary with the prompt, domain, or user preference (Wang et al., 2024). *Geometric constraints* regularize the reward space so dimensions stay aligned with their intended semantic references rather than drifting into arbitrary directions (Jin et al., 2025).

> ❹ Rubric-based optimization should **delay scalar collapse** and preserve **dimension-level structure**, hard constraints, and objective trade-offs.

Grader Analysis. Beyond the reward format and aggregation rule, grader choice is central in direct-reward RL. The grader defines the reward landscape, so its errors are not merely evaluation noise but potential optimization targets. Empirical studies show that downstream gains vary with the judge model, while professional domains further require grounding, domain knowledge, and stable interpretation of fine-grained criteria (Gunjal et al., 2026; Wang et al., 2025c). This creates a cost–robustness trade-off: stronger graders are more expensive, whereas cheaper graders are easier to exploit. Practical systems therefore calibrate graders against stronger references, such as expert judgments, retrieved evidence, or order-swapped comparisons, and may co-evolve the grader with the rubric generator as policy outputs shift during RL (Xu et al., 2026a). Stress tests, including adversarial probes, failure memory, and disagreement checks, further expose underspecified criteria before they become reward shortcuts (Xu et al., 2026d). The goal is not a universally best grader, but a reliability range within which a grader can safely serve as an optimization signal.

Agent Harness. Agentic settings extend credit assignment from final outputs to execution processes. An agent harness records trajectories, tool calls, evidence selection, intermediate artifacts, and subgoal completion, allowing rubric or checklist rewards to supervise process quality rather than only final answers. For deep research agents, process-aware supervision is often more informative than final-answer scoring, because search quality, evidence use, planning, and report construction determine success before the final response is written (Hu et al., 2025; Tongyi DeepResearch Team et al., 2025). For multi-turn tool use, checklist rewards can be attached to turns or subgoals, making failures easier to localize and optimize (Zhang et al., 2026f). Thus, agentic rubric supervision extends PRM-style credit assignment from reasoning steps to tool calls, action sequences, and long-horizon trajectories (Xu et al., 2025).

❺ Rubrics are effective for agentic workflows by **localizing process failures**, but require **high-fidelity traces** rather than visible proxies alone.

4.4 Rubrics for Advanced Training

Beyond scalar rewards, rubrics can serve as training-time control structures that turn evaluation criteria into mechanisms for capability scheduling, failure adaptation, and exploration guidance. We organize these uses into three directions: *curriculum learning*, which stages capability acquisition through criteria; *self-evolving learning*, which updates rubrics from observed failures; and *hint-based learning*, which converts violated criteria into actionable generation guidance.

❹ Rubrics shape model behavior by structuring the **learning trajectory**, including staged objectives, failure adaptation, and **corrective guidance**.

Rubrics for Curriculum Learning. Curriculum learning uses rubrics to schedule optimization objectives rather than only training examples. Because criteria differ in difficulty, priority, and domain specificity, they naturally support staged learning from basic requirements to specialized capabilities (Chen et al., 2026b). Existing work applies this idea across *rubric dimensions*, where perception and constraint following precede harder reasoning; *user preferences*, where personalized reward models progress from general preference recognition to user-specific standards through rubric-like evaluation chains (Zhang et al., 2026c); and *failure taxonomies*, where model errors guide later training toward recurring technical mistakes in domain-adaptive reasoning (Sanders et al., 2026). Overall, rubrics shift curriculum design from ordering data instances to ordering the criteria, capabilities, and failure modes emphasized during optimization.

Rubrics for Self-Evolving Learning. Self-evolving learning treats rubrics as mutable training signals that adapt with the policy. As policy improvement shifts the error distribution, fixed rubrics may miss emerging weaknesses; online methods therefore use rollouts to surface new failures, rewrite them into criteria, and guide later updates (Rezaei et al., 2025). Existing systems instantiate this loop by anchoring or refining rubric sets, retaining self-generated rubrics correlated with correctness, or evolving deep-research criteria alongside the policy (Huang et al., 2025; Shao et al., 2025). This adaptivity is useful when the full criterion set is unknown in advance, but it also introduces feedback-drift risk: rubric generation may start tracking policy artifacts rather than stable task requirements.

Rubrics for Hint-based Learning. Hint-based learning turns violated criteria into actionable guidance, moving rubrics from post-hoc evaluation to generation-time supervision. When exploration is difficult, scalar rewards only indicate failure, whereas rubric-derived hints can localize missing evidence, unsupported reasoning, constraint violations, or unverified visual facts (Zhou et al., 2025; Bi et al., 2025). Existing methods use such hints as reasoning scaffolds, rollout revision signals, self-written success standards (Yu et al., 2026a), or plan-and-verify checklists for multimodal judging (Liu et al., 2026b). Across these variants, rubrics provide operational feedback that makes improvement directions more concrete than final rewards alone.

5 Inference-time Rubric Supervision

Rubrics can also be generated or adapted during inference, serving as dynamic scaffolds for decomposing, judging, and refining candidate outputs when no single ground-truth answer is available. Recent methods derive task-specific supervision from rollouts, critiques, generated principles, pseudo-references, or binary checks, turning uncertain open-ended generation into structured feedback (Bhaskar et al., 2025).

❺ Inference-time rubrics are useful when feedback must be **constructed dynamically**, especially for open-ended outputs requiring **decomposition** and refinement.

The distinction from training-time rubric RL is both temporal and functional: rather than optimizing a policy against fixed rubric rewards, inference-time methods allocate additional computation to construct, refine, or aggregate the rubric signal itself. This includes *reward-guided thinking*, which elicits reasoning before answering; *scalable reward judging*, which samples and aggregates principles or critiques to improve reward quality (Liu et al., 2025b); and *reference-free supervision*, which synthesizes rollouts into pseudo-references and converts them into self-proposed rubric checks (Jayalath et al., 2026). In this way, inference-time rubric supervision connects test-time scaling with post-training by turning rubrics from fixed reward specifications into adaptive scaffolds for evaluating and improving responses at deployment time.

6 Rubric Reward Hacking Risks

Rubrics make open-ended objectives more structured, but they also create new reliability boundaries: optimized criteria and graders can become exploitable proxies. Their scores may remain biased, incomplete, or misaligned with human intent. For example, judges can be sensitive to score-option order, treating rubric levels as multiple-choice candidates rather than semantic anchors (Xu et al., 2026e), and may weight rubric dimensions differently from human evaluators when preferences depend on context, workflow, or implicit expectations (Mittal et al., 2026). Such biases are not reward hacking by themselves, but once optimized, they define an exploitable reward surface.

This risk arises from both the grader and the rubric. On the grader side, weak or biased judges may reward criteria that are only partially satisfied, implicitly inferred, or matched at the wrong specificity level, increasing proxy scores without gains under stronger reference evaluation (Mahmoud et al., 2026). On the rubric side, subjective, non-atomic, underspecified, redundant, low-signal, or directly hackable criteria can let models satisfy the checklist while missing the underlying objective (Qi et al., 2026). Robust rubric-based RL therefore requires stronger graders together with rubric diagnostics, calibration against human or reference-panel judgments, bias and consistency checks, and iterative criterion revision as policy behaviors change. Rubrics should be treated as measurement instruments whose reliability is monitored throughout evaluation and optimization, rather than as fixed reward specifications.

❹ Structured supervision is safe only when rubrics preserve **task-relevant structure** and remain **calibrated**; otherwise, criteria or graders become exploitable proxies.

7 Evaluation

As summarized in Table 4, rubric-based benchmarks provide structured interfaces for comparing subjective, long-form, multimodal, and agentic outputs. Across existing benchmarks, rubrics serve three roles: (1) ***Decomposing*** open-ended quality into checkable criteria, (2) ***Delegating*** judgment to LLM judges, non-LLM verifiers, or hybrid evaluators, and (3) ***Aggregating*** judgments into indicators such as rubric scores, accuracy, success rates, win rates, F1, or risk proportions. We group them into two broad families: (1) ***Real-world task*** benchmarks, where rubrics encode domain or workplace standards, and (2) ***General capability*** benchmarks, where rubrics diagnose agentic behavior, reasoning coverage, alignment, safety, and judge reliability.

7.1 Real-world Tasks

Real-world task benchmarks ask whether model outputs satisfy domain-specific standards rather than generic correctness alone. They span (1) ***Medical*** settings, where rubrics test clinical safety, correctness, and inquiry behavior; (2) ***Legal*** settings, where rubrics assess issue coverage, legal reasoning, and drafting compliance; (3) ***Office Labor*** settings, where rubrics evaluate professional deliverables and workplace execution; (4) ***Academic*** settings, where rubrics measure scientific reasoning, artifact generation, and research reproduction; (5) ***Deep Research*** settings, where rubrics assess evidence grounding, factuality, and research process; and (6) ***Creative Generation*** settings, where rubrics evaluate writing quality and multimodal artifact fidelity.

Medical. Rubric-based medical benchmarks first assess single-response quality in health consultation: physician-written criteria decompose answers into safety, correctness, and communication checks, which LLM evaluators aggregate into weighted

Paper	*Domain*	*Training Set*	*Eval Target*	*Date*	*Rubric Source*	*Modality*	*Indicators*	*# Instances*
				I. Verifier Only				
AgentBoard (Ma et al., 2024)	Agentic	×	Agent Policy	2024-01	Human	Text	Success Rate	1,013
MathCheck (Zhou et al., 2024)	Reasoning	×	Policy Model	2024-07	Hybrid	Text; Visual	Accuracy	4,536
IFBench (Pyatkin et al., 2025)	Alignment	×	Policy Model	2025-07	Human	Text	Accuracy	300
JudgeBench (Tan et al., 2024)	Alignment	×	Judge Model	2024-10	Hybrid	Text	Accuracy	350
				II. LLM as a Judge				
HealthBench (Arora et al., 2025)	Medical	×	Policy Model	2025-05	Human	Text	Rubric Score	5,000
LEGIT (Lee et al., 2026)	Legal	×	Policy Model	2025-11	Hybrid	Text	Rubric Score	24,406
LH-Bench (Gupta and Chandwani, 2026)	Office Labor	×	Agent Policy	2026-03	Human	Text; Visual	Rubric Score	216
TechImage-Bench (Ni et al., 2025)	Academic	×	Policy Model	2025-12	Hybrid	Text; Visual	Rubric Score	654
MiroEval (Ye et al., 2026)	DeepResearch	×	Agent Policy	2026-03	Model	Text; Visual	Rubric Score	100
UEval (Li et al., 2026a)	Creative	×	Policy Model	2026-01	Hybrid	Text; Visual	Rubric Score	1,000
AskBench (Zhao et al., 2026)	Agentic	×	Agent Policy	2026-02	Model	Text	Accuracy	800
MoReBench (Chiu et al., 2025)	Reasoning	×	Policy Model	2025-10	Human	Text	Rubric Score	1,000
InFoBench (Qin et al., 2024)	Alignment	×	Policy Model	2024-01	Human	Text	Accuracy	500
CM2 (Zhang et al., 2026f)	Agentic	✓	Agent Policy	2026-02	Model	Text	Rubric Reward	8,000
mR3 (Anugraha et al., 2026)	Alignment	✓	Judge Model	2025-10	Hybrid	Text	Rubric Reward	100,000
				III. Hybrid Evaluation Method				
FrontierScience (Wang et al., 2026)	Academic	×	Policy Model	2026-01	Human	Text	Rubric Score	160
DEER (Han et al., 2025)	DeepResearch	×	Policy Model	2025-12	Human	Text	Rubric Score	50
TRAJECT-Bench (He et al., 2025a)	Agentic	×	Agent Policy	2025-10	Hybrid	Text	Rubric Score	5,670
RLCER (Sheng et al., 2026)	Reasoning	✓	Policy Model	2026-02	Model	Text	Rubric Reward	17,000

Table 4: Representative rubric-related evaluation resources grouped by evaluation method. The table reports each resource's domain, whether it is used for training, its evaluation target, release date, rubric source, modality, evaluation indicator, and number of instances. Evaluation-method categories are shown as gray full-width group rows. The full paper-level table is provided in Table 7.

rubric scores (Arora et al., 2025), while caregiver-facing advice is audited through clinician-validated risk flags and reported as risk proportions (Goel et al., 2026). Online and multi-turn consultation benchmarks extend this setting to key-point coverage, evidence grounding, risk control, and medical instruction following, using LLM-based rubric scoring or binary test-point accuracy (Wu et al., 2026b; Yan et al., 2026; Yang et al., 2026a). Agentic clinical benchmarks further shift the target from answer quality to inquiry behavior, scoring whether doctor agents ask necessary questions, collect relevant information, and advance diagnosis (Gong et al., 2026).

Legal. Rubric-based legal evaluation first focuses on judicial reasoning, where issue-tree rubrics decompose court-case responses into issue coverage, issue correctness, and final-order correctness, with LLM evaluators aggregating them into rubric scores (Lee et al., 2026). Practice-oriented benchmarks broaden the setting to public consultation, case analysis, and document generation, where point-based rubrics assess issue identification, legal reasoning, statutory application, and drafting compliance (Shi et al., 2026).

Office Labor. Office-labor benchmarks first score final professional deliverables with LLM judges, expert rubrics, weighted checkpoints, or checklist matching, covering professional reports, finance/legal reasoning, complex outputs, and long-form expert generation (Wang et al., 2025c; Akyürek et al., 2025; Liu et al., 2026d; Ruan et al., 2026). Evaluation then moves to workplace realism: domain experts compare model work products against human-expert deliverables and report win rates (Patwardhan et al., 2025); customer-service and freelance-marketplace tasks evaluate service quality, tool use, risk, hallucination, and job completion through LLM or human scoring (Gao et al., 2025; Yi et al., 2025). Long-horizon enterprise and high-value professional benchmarks further assess intermediate artifacts, final deliverables, and economically consequential tasks with human-anchored or expert rubrics and rubric scores (Gupta and Chandwani, 2026; Yang et al., 2026b).

Academic. Academic benchmarks progress from scientific reasoning to research execution. Expert-level science tasks combine verifier-like final-answer checking for olympiad-style problems with human-anchored rubrics and LLM judges for research-level subtasks, reporting accuracy or rubric scores (Wang et al., 2026). Technical artifact benchmarks extend evaluation to scientific and engineering image generation, where hybrid rubrics assess technical accuracy, structural completeness, information coverage, and source faithfulness (Ni et al., 2025). More agentic settings evaluate full paper reproduction, where agents implement methods, run experiments, and match reported results using hierarchical or hybrid rubric scoring (Starace et al., 2025; Qiu et al., 2026a).

Deep Research. Deep-research benchmarks progress from final-report scoring to evidence- and process-aware assessment. Early benchmarks use expert-written rubrics and LLM judges to score evidence use, factual grounding, reasoning quality, structure, and instruction following (Sharma et al., 2025). Evidence-aware benchmarks add citation grounding, retrieval support, and report coverage checks through numeric or checklist-style rubric scores (Du et al., 2025; Li et al., 2026b). More recent benchmarks combine LLM-as-a-judge scoring with claim-level evidence verification (Han et al., 2025), while process-aware multimodal evaluation further scores how agents search, reason, synthesize evidence, handle text–vision inputs, and refine analysis (Ye et al., 2026).

Creative Generation. Creative-generation benchmarks progress from text quality to multimodal artifact fidelity. Text-only settings evaluate long-form writing with instance-specific or hybrid checklist/rubric criteria, using LLM judges to score final response quality and human-aligned dimensions (Wu et al., 2025; Que et al., 2024). Beyond text, presentation-generation benchmarks use human-anchored binary rubrics and LLM/MLLM evaluators to assess slide content, structure, and visual presentation (Chen et al., 2026a). Unified multimodal generation extends this setting to mixed image–text outputs, where MLLM evaluators apply binary rubrics to assess visual and textual requirements (Li et al., 2026a).

7.2 General Capability Evaluation

While real-world benchmarks emphasize domain standards, general capability benchmarks diagnose model capabilities and evaluator reliability beyond final answers. They span (1) ***Agentic*** evaluation, where rubrics track progress, tool use, trajectories, long-term user context, and clarification behavior; (2) ***Reasoning*** evaluation, where rubrics test whether models cover key intermediate considerations rather than only final correctness; and (3) ***Alignment*** evaluation, where rubrics assess instruction following, safety, judge reliability, and rubric-generator quality.

Agentic. Agentic benchmarks evaluate how models complete interactive tasks and progress through multi-step execution. Early settings cover embodied, game, web, and tool-use environments, where human-anchored criteria are checked by environment- or verifier-style evaluators and summarized by success rates and progress metrics (Ma et al., 2024). Tool-use and trajectory-diagnostic benchmarks extend this to real MCP servers and API-based multi-tool calling, using non-LLM or hybrid evaluators to assess task success, tool selection, arguments, dependency ordering, and trajectory-level rubric scores (Luo et al., 2025; He et al., 2025a). Other settings add persistent user context or clarification behavior, reporting task success or accuracy-style clarification metrics through verifier-based evaluation or LLM judges (Xiu et al., 2026; Zhao et al., 2026).

Reasoning. Reasoning benchmarks evaluate both final correctness and coverage of intermediate considerations. In moral reasoning, human-anchored rubrics decompose responses into values, stakeholder trade-offs, harm avoidance, and actionable recommendations; LLM evaluators judge criterion coverage and report rubric or reasoning scores (Chiu et al., 2025). In mathematical reasoning, checklist-style tests cover problem solving, answerability, outcome verification, and faulty reasoning across text and geometry settings, using non-LLM verifiers or answer-matching rules to report accuracy and related checklist metrics (Zhou et al., 2024).

Alignment. Alignment benchmarks evaluate policy compliance, safety, and evaluator reliability. Instruction-following settings decompose final responses into requirement-level or verifiable constraints, using LLM evaluators or non-LLM verifiers to report binary satisfaction and accuracy (Qin et al., 2024; Pyatkin et al., 2025). Multilingual, multi-turn, system-level, and context-heavy variants further test constraint preservation across languages and conversations with rule-based, rubric-verifier, or LLM-as-a-judge pipelines (Li et al., 2025b; He et al., 2024, 2025b; Sirdeshmukh et al., 2025). Safety-oriented benchmarks use LLM rubric-based evaluators to score harmful compliance (Souly et al., 2024). Beyond policy outputs, judge and reward-model benchmarks test pairwise, rubric-level, multilingual, and research-novelty judgments against gold or human-anchored labels, reporting accuracy, Macro-F1, preference accuracy, or error metrics (Tan et al., 2024; Pan et al., 2026; Whitehouse et al., 2026; Schopf and

Färber, 2026). Rubric-level benchmarks further evaluate generators through rubric recall, hallucination, structural F1, and preference recovery (Zhang et al., 2026e).

➎ Rubrics supervise **subjective or agentic evaluation** by turning standards into checkable criteria, evaluator judgments, and **comparable indicators**.

8 Application

Rubrics act as task interfaces that turn open-ended goals into explicit, checkable criteria. Across applications, they serve three recurring roles: (1) ***Evaluation Criteria*** for assessing policy, agent, or judge outputs, (2) ***Reward Signals*** for optimizing policies, agents, or reward models, and (3) ***Supervision Signals*** for fine-tuning, alignment, or self-revision. This framing highlights how rubrics bridge domain-specific requirements with actionable supervision, making evaluation and optimization more interpretable across diverse settings. Table 5 summarizes representative applications across domains, agentic workflows, and multimodal settings.

Medical. Rubrics make clinical safety and correctness operational. Medical responses must satisfy factual correctness, safety, caution, and communication requirements simultaneously. Rubrics decompose these clinical norms into explicit criteria, allowing them to serve as RL rewards, alignment supervision, or evaluation signals. In medical RL and dialogue training, rubric scores guide models toward safer and more complete answers (Baichuan-M2 Team et al., 2025; Wang et al., 2025b; Yang et al., 2026c). Physician-verified rubrics and principles further support clinician alignment and self-revision (Lyu et al., 2026a), while medical QA and dental visual reasoning use rubrics to encode diagnostic correctness, risk control, and tool-use requirements (Xu et al., 2026b; Fan et al., 2026a).

Writing & Retrieval. Rubrics turn subjective quality and grounding into measurable criteria. Long-form explanations and RAG outputs are difficult to judge by binary correctness, since they may be fluent but incomplete, relevant but weakly grounded, or coherent but not useful. Rubrics specify what a good explanation or retrieved answer should contain. They support audience-aware evaluation of well-being explanations (Jiang et al., 2025a), measure relevance and grounding in retrieval/RAG systems (Lan et al., 2025; Farzi and Dietz, 2024), and make broad text-quality dimensions such as coherence, usefulness, and completeness more stable to evaluate (Hashemi et al., 2024).

DeepResearch. Rubrics provide intermediate structure for evidence-grounded long-form research. DeepResearch outputs depend on search, evidence selection, citation grounding, and synthesis, so final-report evaluation alone may miss unsupported claims or shallow evidence use. Rubrics scaffold this process by checking coverage, evidence quality, citation faithfulness, and synthesis. Checklist-style judges provide training rewards for research agents (Hu et al., 2025), citation-aware rubrics connect claims to evidence during deep search (Zhang et al., 2026a), and evolving or query-specific rubrics adapt criteria to each research question (Shao et al., 2025; Lv et al., 2026).

Code. Rubrics complement execution tests with contextual criteria. Execution tests are useful but often incomplete: a patch may pass tests while violating repository conventions, missing edge cases, or failing to satisfy issue intent. Rubrics address these gaps by checking code against contextual and task-specific requirements. Contextual SWE rubrics evaluate patches beyond simple execution (Raghavendra et al., 2026), while CUDA programming combines rubric rewards with execution feedback for specialized code generation (Li et al., 2026c).

General Agentic. Rubrics provide process-level feedback for multi-step agents. Agentic systems can fail through poor planning, wrong tool calls, missing observations, weak recovery, or incorrect final answers. A single success/failure reward gives little credit assignment. Rubrics and checklists decompose trajectories into intermediate criteria, enabling denser and more interpretable feedback. Checklist rewards guide multi-step tool use (Zhang et al., 2026f), task-adaptive rubrics score trajectories and synthesize rewards (Ding, 2026), and synthetic tool-use environments use rubric rewards for agent training (Lyu et al., 2026b). Structured rubrics also support tool-use reward modeling, long-horizon alignment, and process-aware RL (Jiang and Ferraro, 2026; Masters et al., 2025; Zhang et al., 2026d).

Paper	*Application Setting*	*Date*	*Rubric Role*	*Eval Target*	*Rubric Source*	*Modality*	*Eval Method*
I. Medical							
Baichuan-M2 (Baichuan-M2 Team et al., 2025)	Medical	2025-09	Reward Signal	Policy Model	Model-generated	Text	LLM
ClinAlign (Lyu et al., 2026a)	Medical	2026-02	Supervision Signal	Policy Model	Hybrid	Text	LLM
II. Writing & Retrieval							
LLM-Rubric (Hashemi et al., 2024)	Writing	2024-12	Evaluation Criteria	Policy Model	Human-anchored	Text	LLM
Retro* (Lan et al., 2025)	Retrieval	2025-09	Reward Signal	Retriever	Human-anchored	Text	LLM
III. DeepResearch							
CaRR (Zhang et al., 2026a)	DeepResearch	2026-01	Reward Signal	Agent Policy	Model-generated	Text	LLM
Query-Specific (Lv et al., 2026)	DeepResearch	2026-02	Reward Signal	Rubric Generator	Hybrid	Text	LLM
IV. Code							
Agentic Rubrics (Raghavendra et al., 2026)	Code	2026-01	Evaluation Criteria	Agent Policy	Model-generated	Text	LLM
StitchCUDA (Li et al., 2026c)	Code	2026-03	Reward Signal	Agent Policy	Hybrid	Text	Hybrid
V. General Agentic							
CM2 (Zhang et al., 2026f)	General Agentic	2026-02	Reward Signal	Agent Policy	Model-generated	Text	LLM
AdaRubric (Ding, 2026)	General Agentic	2026-03	Reward Signal	Agent Policy	Model-generated	Text	LLM
VI. Multimodal							
RubricRL (Feng et al., 2025)	Text + Vision	2025-11	Reward Signal	Policy Model	Model-generated	Text + Vision	LLM
SpeechLLM FT (Parikh et al., 2026)	Text + Audio	2026-03	Supervision Signal	Policy Model	Human-anchored	Text + Audio	LLM
Omni-RRM (Kong et al., 2026)	Omni-modal	2026-02	Reward Signal	Judge Model	Model-generated	Omni	LLM

Table 5: Representative rubric-based applications grouped by application setting. The table summarizes how rubrics are used across domains and modalities, including each paper's release date, rubric role, evaluation target, rubric source, modality, and evaluation method. Application-setting categories are shown as gray full-width group rows. The full paper-level table is provided in Table 8.

8.1 Multimodal Applications

Text + Vision. Rubrics decompose visual quality into semantic, spatial, and perceptual dimensions. Visual outputs may look plausible while missing objects, misplacing attributes, hallucinating details, or failing to follow the prompt. Rubrics make these errors diagnosable by checking object correctness, attribute accuracy, spatial consistency, OCR fidelity, caption completeness, and reasoning faithfulness. They serve as rewards for text-to-image generation (Feng et al., 2025), supervise dense captioning (Huang et al., 2026), and improve VLM reward modeling or visual preference optimization (Qiu et al., 2026b; Tian et al., 2026; Yu et al., 2026b). Rubric-based generative rewards also help evaluate multimodal reasoning beyond final answers (Jia et al., 2025).

Text + Audio. Rubrics separate speech assessment into interpretable dimensions. Speech quality involves pronunciation, fluency, prosody, intelligibility, rhythm, and expressiveness, which cannot be captured well by a single holistic score. Rubric-guided SpeechLLM fine-tuning uses these dimensions as structured supervision (Parikh et al., 2026), while speech judge models use structured criteria to explain and assess speech quality (Wang et al., 2025a).

Omni-modal. Rubrics provide a shared evaluation language across heterogeneous modalities. Omni-modal systems must compare text, image, video, and audio outputs under a unified preference framework. Rubric-grounded reward modeling provides structured criteria for cross-modal preference judgments (Kong et al., 2026). This shared interface is especially promising for multimodal role-playing, embodied assistants, and personalized interactive agents, where outputs must satisfy persona consistency, grounding, safety, memory consistency, and long-term coherence across diverse interaction contexts and modality-specific generation constraints.

> ❺ Rubrics provide **structured supervision** by turning domain, agentic, and multimodal requirements into **actionable signals** for assessment, optimization, and alignment.

9 Future Directions

Rubric-centered research is moving from using rubrics as static evaluation prompts toward treating them as structured supervision interfaces for evaluation, reward modeling, and alignment. We outline future directions from two complementary perspectives: methodological advances in rubric-centered supervision and application-oriented extensions to broader AI systems.

Methodological Level. *Scalable and Reliable Rubric Construction.* A central challenge is to balance expert fidelity with automatic scalability.

Expert-written rubrics provide strong domain validity but are costly to obtain, whereas model-generated rubrics are scalable but may be vague, redundant, incomplete, or difficult to verify. Future construction pipelines should better integrate expert knowledge, preference data, references, retrieved evidence, and failure patterns to produce criteria that are specific, discriminative, reusable, and auditable.

Robust Rubric-based Reward Signals. When rubrics are used for reward modeling or reinforcement learning, criterion quality directly affects policy optimization. Future work should improve criterion selection, weighting, redundancy control, aggregation, and credit assignment under changing prompts, tasks, models, and optimization pressure. In particular, rubric-based rewards should preserve dimension-level structure before scalarization, so that critical constraints are not averaged away and weak criteria do not become exploitable shortcuts.

Adaptive Rubric–Judge–Policy Optimization. Rubrics should be optimized together with judges, policies, and verifiers rather than treated as fixed prompts. As model capabilities improve, static rubrics may lose discriminative power or become easy to game. Curriculum learning, self-evolving rubrics, and online failure mining provide promising mechanisms for adapting criteria to task difficulty and observed errors. However, such adaptation also introduces feedback-drift risks, where rubrics overfit policy artifacts instead of stable human objectives. Distinguishing useful rubric evolution from objective drift remains an important problem.

Meta-evaluation and Interpretability. Future benchmarks should evaluate not only model outputs, but also rubric quality, judge faithfulness, verifier coverage, and the causal relationship between rubric satisfaction and task success. A rubric can be consistent and easy to apply while still failing to capture the factors that matter for downstream utility. More fine-grained meta-evaluation is therefore needed to test whether criteria are executable, whether judges apply them faithfully, and whether rubric-based optimization improves real performance rather than merely increasing proxy scores.

Application Level. *Agentic and Long-horizon Supervision.* Rubrics should extend beyond final-answer scoring to planning, tool use, information seeking, memory use, and multi-step agentic workflows. In these settings, failures often arise from intermediate decisions rather than final responses alone. Future work should develop process-aware rubrics and high-fidelity agent harnesses that support reliable trajectory capture, failure localization, and credit assignment.

Personalized and Context-aware Alignment. Personalized rubrics and user-conditioned reward models can adapt evaluation criteria to different users, domains, and interaction contexts while retaining interpretability. The main challenge is to reconcile context-specific preferences with stable alignment principles. Future systems should reflect user intent and domain norms without weakening general requirements such as factuality, safety, fairness, and robustness.

Multimodal and Domain-grounded Rubrics. Future rubrics should better connect natural-language criteria with visual evidence, executable tests, structured databases, and domain-specific validators. For multimodal tasks, an important direction is to distinguish criteria that transfer across modalities from those requiring modality-specific evidence, such as spatial relations, temporal coherence, or prosodic quality. For professional domains, rubrics should be grounded in external knowledge, expert standards, and executable verification rather than surface-level textual matching.

Governance and Safety. As rubrics enter data construction, reward modeling, and reinforcement learning pipelines, their errors may be amplified and internalized into model behavior. Automatic rubric generation should therefore be paired with human review, evidence locking, uncertainty reporting, version control, bias diagnostics, and adversarial testing. Rubrics should be treated as measurement instruments that require continuous monitoring and calibration, not as fixed reward specifications.

Overall, these directions point toward rubrics as adaptive supervision objects: constructed from data, validated by experts and evaluators, optimized for learning, and updated as models, tasks, and users evolve.

10 Conclusion

This survey presents a unified view of rubrics as a structured supervision paradigm for complex AI systems. By externalizing expert judgment into explicit and machine-operational criteria, rubrics provide an effective interface between human objectives and scalable model-based evaluation. We organize existing studies across six dimensions: data construction, training integration, inference enhancement, risks, evaluation mechanisms, and applications, showing how rubrics are evolving from static evaluation templates into reusable, adaptive, trainable, and auditable supervision signals. We hope this survey offers a concise foundation for future research on rubric-centered evaluation, alignment, and post-training, and highlights rubrics as a promising path toward scalable, interpretable, and controllable supervision.

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A Survey Scope and Positioning

A.1 Motivation and Necessity of This Survey

The motivation for this survey stems from three observations about the changing role of evaluation and feedback in foundation models. First, model development is moving from single-turn response generation toward open-ended, professional, multimodal, and agentic tasks, where success depends on multiple objectives rather than one final answer. As discussed in the introduction, modern systems must satisfy correctness, safety, planning quality, evidence use, policy compliance, interaction quality, robustness, and cost constraints. Conventional signals such as references, scalar ratings, pairwise preferences, and exact verifiers compress these requirements into coarse labels, making it difficult to explain which constraint is satisfied, violated, or missing.

Need for Structured Supervision. Rubrics address this gap by decomposing human objectives into explicit, inspectable, and reusable criteria. This is especially necessary when evaluation itself becomes part of the training loop: rubric criteria may guide data annotation, judge behavior, reward modeling, reinforcement learning, inference-time revision, and error diagnosis. Without a unified account of how rubrics are constructed and used, the same object may be treated inconsistently as a checklist, scoring guide, judge prompt, reward specification, or process constraint.

Fragmentation Across Research Threads. Existing rubric-related studies are spread across data construction, LLM judging, preference learning, direct rubric rewards, process supervision, agent benchmarks, multimodal evaluation, and domain-specific alignment. The paper therefore needs a survey that follows rubrics across their full lifecycle: from expert or model-based construction, to criterion-level judging, to aggregation, optimization, deployment, and risk analysis. This lifecycle view is necessary because rubric errors can propagate downstream: vague, incomplete, or incorrect criteria may mislead judges, distort rewards, or create reward-hacking opportunities.

Lack of Shared Design Vocabulary. The survey is also motivated by the need to clarify what makes rubric supervision reliable. The preliminaries identify principles such as atomicity, context specificity, checkability, completeness, and correctness. These principles provide a vocabulary for comparing rubrics beyond task performance alone, and help explain why rubrics are useful for auditing both evaluation outcomes and optimization targets.

A.2 Focus and Scope Delimitation

This survey focuses on rubrics as a middle-layer representation between implicit human judgment and machine-operational supervision. We use *rubric* to mean an explicit set of criteria that can be judged at the criterion level, aggregated into evaluation signals, reused across examples, or converted into training feedback. The survey primarily covers foundation-model settings where rubrics play an operational role in evaluation, alignment, post-training, inference-time guidance, or domain deployment.

Included Scope. We organize the literature according to the taxonomy used in the main paper. The data-construction scope includes expert-based annotation, model-based generation, preference-assisted induction, iterative refinement, and human–AI collaboration. The training scope includes rubric generators and judges, preference-reward learning, direct rubric-reward RL, dense or token-level reward construction, process-level supervision, multi-objective aggregation, self-evolving rubrics, and rubric-guided exploration. The evaluation scope includes resources where rubrics assess final answers, reasoning steps, tool-use trajectories, professional decisions, multimodal outputs, or agent behaviors. The application scope covers real-world and general-capability settings, including medicine, law, office labor, academic work, deep research, creative generation, code, agentic workflows, vision-language tasks, audio, and omni-modal systems.

Excluded Scope. We do not attempt to review all evaluation or alignment methods. Benchmark leaderboards, preference datasets, reward models, verifiers, and LLM-judge protocols are included only when explicit rubric criteria are central to the method or resource. We also do not treat educational assessment rubrics as an independent historical field, except when they help explain transparency and scoring consistency. Finally, optimization algorithms are not the organizing unit of this survey; they are discussed only when rubrics specify, decompose, adapt, or diagnose the supervision signal being optimized.

B Comparison with Existing Surveys

While our survey establishes a unified taxonomy for *rubric-centered supervision* in foundation models, it is important to distinguish its scope from two closely related lines of surveys: LLM-as-Judge and RL for LLMs. Existing surveys in these areas provide valuable perspectives on scalable evaluation and post-training optimization. However, they primarily focus on either the evaluator model or the optimization algorithm. In contrast, our survey focuses on the intermediate supervision representation that connects human judgment, model-based evaluation, and policy optimization: rubrics.

LLM-as-Judge. Surveys on LLM-as-Judge mainly study how large language models can serve as evaluators for open-ended outputs, including pairwise comparison, scalar scoring, critique generation, and benchmark evaluation (Li et al., 2024). Their central questions include how to prompt or train judge models, how to design scoring protocols, and how to improve agreement with human preferences (Gu et al., 2025). Recent work further emphasizes judge reliability, covering issues such as position bias, verbosity bias, self-preference, calibration errors, and meta-evaluation (Li et al., 2025a). These studies are fundamental to rubric-based evaluation because LLM judges are often used to apply rubric criteria at scale. However, their primary unit of analysis is the *judge*: how an evaluator produces scores, rankings, critiques, or decisions. Our survey shifts the focus from the evaluator itself to the *structured criteria* being evaluated. In rubric-centered supervision, an LLM judge is only one possible executor, alongside human experts, reward models, programmatic verifiers, and hybrid evaluators. The key question is not only whether a judge is reliable, but also what criteria it is asked to judge, how these criteria are constructed, whether they are atomic and checkable, how criterion-level judgments are aggregated, and how rubric errors propagate into downstream optimization. Thus, while LLM-as-Judge surveys mainly ask how models can evaluate, our survey asks how human objectives should be decomposed into explicit, reusable, and optimizable evaluation criteria.

RL for LLMs. Surveys on RL for LLMs mainly investigate how reinforcement learning and preference optimization improve language model behavior during post-training (Anonymous, 2025). They typically organize the literature around optimization paradigms and algorithms, such as RLHF, PPO, DPO, GRPO, and reward-guided test-time scaling (Wu, 2025). Their central concern is how to learn from reward signals efficiently and stably, how to reduce data or rollout cost, and how different policy optimization objectives affect reasoning, alignment, and generalization (Yu et al., 2026c). Preference-learning surveys further analyze how human preferences are collected, modeled, and optimized for alignment (Jiang et al., 2025b). These surveys provide the algorithmic background for rubric-guided training, especially when rubric scores are used as rewards in GRPO-style or direct-reward RL. Nevertheless, their focus is usually on how to optimize a policy given a reward, whereas our survey focuses on *how the reward itself is specified, decomposed, judged, aggregated, and adapted*. Rubrics are not merely another reward source; they expose the internal structure of the objective by separating dimensions such as correctness, factuality, safety, completeness, evidence use, tool behavior, and process quality. This perspective highlights issues that are often secondary in general RL surveys, including rubric construction, criterion granularity, grader reliability, multi-objective aggregation, credit assignment, self-evolving rubrics, and rubric reward hacking. Therefore, while RL-for-LLMs surveys ask how models learn from rewards, our survey asks how structured rewards can be designed so that what models learn remains interpretable, auditable, and aligned with human intent.

Work	*Date*	*Opt. Target*	*Opt. algorithm*	*Jud. Format*	*Main Train Dataset*	*Modality*	*Focus*
SFT							
R3 (Anugraha et al., 2025)	2025-05	Reward model	SFT	Numeric	R3-D14k	Text	General reward modeling
P-Check (Seo and Lee, 2026)	2026-01	Rubric generator	SFT	Binary	PRISM-Personalized	Text	Personalization
OpenRubrics (Liu et al., 2025a)	2026-02	Rubric generator	SFT	Holistic	OpenRubrics	Text	Reward modeling
Preference-Reward RL							
RLCF (Viswanathan et al., 2025)	2025-07	Policy model	DPO	Binary	WildChecklists	Text	Instruction following
OpenRS (Jia et al., 2026)	2026-02	Policy model	BRPO	Numeric	Self-construction	Text	Open-ended alignment
Rubric-ARM (Xu et al., 2026a)	2026-02	Reward model	Alt-GRPO	Holistic	OpenRubrics	Text	Rubric generator-judge co-training
AdaRubric (Ding, 2026)	2026-03	Agent policy	DPO	Numeric	Self-construction	Text+Vision	Agent trajectory eval.
C2 (Kawabata and Sugawara, 2026)	2026-04	Reward model	Rubric-aug. RM	Binary	Preference pairs	Text	Binary-preference reward modeling
Visual Preference Opt.	2026-04	Policy model	VPO	Numeric	Self-construction	Text+Vision	Visual preference optimization
ROPD (Fang et al., 2026)	2026-05	Policy model	OPD	Numeric	DAPO-Math-17K	Text	Knowledge distillation
Rubric-Based Grader							
RaR (Gunjal et al., 2026)	2025-07	Policy model	GRPO	Holistic	RaR datasets	Text	Med./science reasoning
RuscaRL (Zhou et al., 2025)	2025-08	Policy model	GRPO	Binary	HealthBench etc.	Text	General reasoning
Search-Gen-V (Ma et al., 2025)	2025-10	Grader	Rubric verification	Ternary	Search-augmented QA	Text	Search-augmented verification
RULERS (Hong et al., 2026)	2026-01	Grader	Evidence-anchored scoring	Binary	Self-construction	Text	Robust evidence-grounded judging
AdvancedIF (He et al., 2025b)	2025-11	Policy model	RIFL	Binary	Self-construction	Text	Rubric IF training
RuRL (Chen et al., 2026b)	2026-01	Policy model	DAPO	Binary	RubricHub	Text	Open-ended alignment
ClinAlign (Lyu et al., 2026a)	2026-02	Policy model	GRPO	Binary	HealthRubrics	Text	Medical alignment
Auto RaR (Tian et al., 2026)	2026-05	Policy model	RPO	Holistic	ShareGPT-4o-Image	Text+Vision	Multimodal generation
Multi-objective Opt.							
ArmoRM (Wang et al., 2024)	2024-06	Reward model	MoE reward modeling	Numeric	Preference data	Text	Context-dependent scalarization
Rubric Anchors (Huang et al., 2025)	2025-08	Policy model	Stage-wise RL	Numeric	Proprietary corpus	Text	Stage-wise rubric anchors
MRRML (Jin et al., 2025)	2025-11	Reward model	Geometric constraint	Numeric	Self-construction	Text	Geometric reward regularization
GDPO (Liu et al., 2026c)	2026-01	Policy model	GDPO	Numeric	ToolACE, Hammar etc.	Text	Multi-reward RL
Quark-Alignment (Xu et al., 2026b)	2026-02	Policy model	Uni-Reward	Numeric	Self-construction	Text	Medical multi-obj. alignment
ARL-RR (Lan et al., 2026)	2026-03	Policy model	Alternating RL	Numeric	Self-construction	Text	Contextual rubric rewards
PAPO (Tan et al., 2026)	2026-03	Policy model	PAPO	Numeric	NuminaMath-1.5-RL-Verifiable	Text	Mathematical reasoning
Credit Assignment							
DRO (Xu et al., 2025)	2025-06	Policy model	GRPO	Binary	ParaRev etc.	Text	Token-level reward with rubric
Miracle Steps (Yuan et al., 2025)	2025-10	Policy model	Rubric reward	Binary	Math reasoning	Text	Reasoning-step correction
Grad2Reward (Zhang et al., 2026g)	2026-02	Policy model	Token-level GRPO	Binary	HealthBench etc.	Text	Dense reward from sparse judgment
Features as Rewards (Prasad et al., 2026)	2026-02	Policy model	RLFR	Numeric	Longfact++ etc.	Text	Feature-level reward
Rubric-Supervised Critic	2026-03	Reward model	Critic training	Numeric	Real-world outcomes	Text	Sparse-outcome credit assignment
Rubrics to Tokens (Xu et al., 2026c)	2026-04	Policy model	RTT-GRPO	Numeric	HiR-16K	Text	Token-level rubric reward
Agent Harness							
Step-DeepResearch (Hu et al., 2025)	2025-12	Agent policy	PPO	Binary	Self-construction	Text	Checklist deep-research RL
TableGPT-R1 (Yang et al., 2025)	2025-12	Agent policy	GRPO++	Numeric	Self-construction	Text	Table reasoning agent RL
ArenaRL (Zhang et al., 2026d)	2026-01	Agent policy	ArenaRL	Holistic	Open-Travel etc.	Text	Intra-group trajectory ranking
CaRR	2026-01	Agent policy	C-GRPO	Binary	Deep-Dive	Text	Citation-aware agent rewards
CM2 (Zhang et al., 2026f)	2026-02	Agent policy	GRPO	Binary	Nemotron-PT-v1 etc.	Text	Process credit for tool use
SWE-TRACE (Han et al., 2026)	2026-04	Agent policy	Agentic RL + TTS	Binary	SWE benchmarks	Text	Rubric PRM for SWE agents
Curriculum Learning							
InfiMed-ORBIT (Wang et al., 2025b)	2025-10	Policy model	GRPO	Numeric	DoctorAgent-RL etc.	Text	Incremental medical alignment
Data-Driven Reasoning	2026-02	Reward model	Rubric generation	Numeric	Domain-adaptive data	Text	Domain-adaptive curricula
RuCL (Chen et al., 2026b)	2026-02	Policy model	GRPO	Numeric	ViRL-39K	Text+Vision	Difficulty-aware scheduling
P-GenRM (Zhang et al., 2026c)	2026-02	Policy model	GRPO	Numeric	PersonalRewardBench etc.	Text	Personalized GenRM
Self-Evolving							
Online Rubrics (Rezaei et al., 2025)	2025-10	Rubric generator	Online elicitation	Binary	Pairwise comparisons	Text	Online rubric evolution
DR Tulu (Shao et al., 2025)	2025-11	Policy model	GRPO	Ternary	SearchArena etc.	Text	Evolving deep-research rubrics
SibylSense (Xu et al., 2026d)	2026-02	Policy model	GRPO	Binary	RaR-Medicine + GovReport	Text	Memory-adaptive rubric rewards
RLCER	2026-02	Policy model	PPO	Binary	DAPO-Math-17K	Text	Self-evolving CoT rubrics
Hint-based Learning							
RuscaRL (Zhou et al., 2025)	2025-08	Policy model	GRPO	Binary	HealthBench etc.	Text	General reasoning
Reward and Guidance (Bi et al., 2025)	2025-11	Policy model	RGR-GRPO	Numeric	WebInstruct-Verify	Text	Rubric-guided exploration
Think-with-Rubrics (Yu et al., 2026a)	2026-05	Policy model	Rubric-guided reasoning	Binary	Self-construction	Text	Internal reasoning guidance
DeltaRubric (Liu et al., 2026b)	2026-05	Reward model	Multi-role RL	Binary	Self-construction	Text+Vision	Plan-and-verify guidance
Reward Hacking							
CROME (Srivastava et al., 2026)	2025-06	Reward model	Causal augmentation	Holistic	RewardBench etc.	Text	Causal robustness to reward hacking
Rubric Position Bias (Xu et al., 2026e)	2026-02	Grader	Balanced permutation	Numeric	MT-Bench etc.	Text	Rubric option position bias
Code Eval Biases (Mittal et al., 2026)	2026-03	Grader	Bias analysis	Holistic	Code evaluation data	Text	Human–LLM judge bias
RIFT (Qi et al., 2026)	2026-04	Rubric diagnostic	Failure-mode diagnosis	Binary	Five benchmark rubrics	Text	Rubric failure diagnostics
Rubric Hacking (Mahmoud et al., 2026)	2026-05	Policy model	GRPO	Binary	RubricHub etc.	Text	Verifier exploitation

Table 6: Representative rubric-related training methods organized by section, optimization target, grading interface, training objective, dataset, modality, and focus. Section labels are shown as gray full-width group rows; works within each section are ordered chronologically.

Paper	*Domain*	*Training Set*	*Eval Target*	*Date*	*Language*	*Rubric Source*	*Modality*	*Indicators*	*# Instances*
I. Verifier Only									
AgentBoard (Ma et al., 2024)	Agentic	×	Agent Policy	2024-01	English	Human	Text	Success Rate	1,013
MCP-Universe (Luo et al., 2025)	Agentic	×	Agent Policy	2025-08	English	Human	Text	Success Rate	231
ASTRA-bench (Xiu et al., 2026)	Agentic	×	Agent Policy	2026-03	English	Human	Text	Success Rate	2,413
MathCheck (Zhou et al., 2024)	Reasoning	×	Policy Model	2024-07	English; Chinese	Hybrid	Text; Visual	Accuracy	4,536
IFBench (Pyatkin et al., 2025)	Alignment	×	Policy Model	2025-07	English	Human	Text	Accuracy	300
JudgeBench (Tan et al., 2024)	Alignment	×	Judge Model	2024-10	English	Hybrid	Text	Accuracy	350
RINoBench (Schopf and Färber, 2026)	Alignment	×	Judge Model	2026-03	English	Human	Text	Accuracy	1,381
RubricEval (Pan et al., 2026)	Alignment	×	Judge Model	2026-03	English	Human	Text	Accuracy	3,486
Multi-IF (He et al., 2024)	Alignment	×	Policy Model	2024-10	8 languages	Hybrid	Text	Accuracy	4,501
II. LLM as a Judge									
LiveMedBench (Yan et al., 2026)	Medical	×	Policy Model	2026-02	English; Chinese	Hybrid	Text	Rubric Score	2,756
QuarkMedBench (Wu et al., 2026b)	Medical	×	Policy Model	2026-03	Chinese	Hybrid	Text	Rubric Score	24,674
RubRIX (Goel et al., 2026)	Medical	×	Policy Model	2026-01	English	Human	Text	Risk Proportion	20,338
MedDialogRubrics (Gong et al., 2026)	Medical	×	Agent Policy	2026-01	English; Chinese	Hybrid	Text	Rubric Score	5,200
MedMT-Bench (Yang et al., 2026a)	Medical	×	Policy Model	2026-03	English	Hybrid	Text; Visual	Accuracy	400
InfiMed-ORBIT (Wang et al., 2025b)	Medical	✓	Policy Model	2025-10	Chinese	Hybrid	Text	Rubric Reward	28,000
HealthBench (Arora et al., 2025)	Medical	×	Policy Model	2025-05	49 languages	Human	Text	Rubric Score	5,000
LEGIT (Lee et al., 2026)	Legal	×	Policy Model	2025-11	Korean	Hybrid	Text	Rubric Score	24,406
PLawBench (Shi et al., 2026)	Legal	×	Policy Model	2026-01	Chinese	Human	Text	Rubric Score	850
PRBench (Akyürek et al., 2025)	Office Labor	×	Policy Model	2025-11	English	Human	Text	Rubric Score	1,100
ProfBench (Wang et al., 2025c)	Office Labor	×	Policy Model	2025-10	English	Human	Text	Rubric Score	80
$OneMillion-Bench (Yang et al., 2026b)	Office Labor	×	Policy Model	2026-03	English; Chinese	Human	Text	Rubric Score	400
XpertBench (Liu et al., 2026d)	Office Labor	×	Policy Model	2026-03	English	Hybrid	Text	Rubric Score	1,346
ExpertLongBench (Ruan et al., 2026)	Office Labor	×	Policy Model	2025-06	English	Human	Text	F1	1,050
OlaBench (Gao et al., 2025)	Office Labor	×	Agent Policy	2025-10	English	Hybrid	Text	Rubric Score	5,000
LH-Bench (Gupta and Chandwani, 2026)	Office Labor	×	Agent Policy	2026-03	English	Human	Text; Visual	Rubric Score	216
TechImage-Bench (Ni et al., 2025)	Academic	×	Policy Model	2025-12	English	Hybrid	Text; Visual	Rubric Score	654
SciRM (Sahinuc et al., 2026)	Academic	✓	Judge Model	2026-01	English	Human	Text	Rubric Reward	58,712
ResearchRubrics (Sharma et al., 2025)	DeepResearch	×	Policy Model	2025-11	English	Human	Text	Rubric Score	101
DeepResearch Bench (Du et al., 2025)	DeepResearch	×	Policy Model	2025-06	English; Chinese	Hybrid	Text	Rubric Score	100
MiroEval (Ye et al., 2026)	DeepResearch	×	Agent Policy	2026-03	English; Chinese	Model	Text; Visual	Rubric Score	100
DeepResearchBenchII (Li et al., 2026b)	DeepResearch	×	Policy Model	2026-01	English; Chinese	Hybrid	Text	Rubric Score	132
UEval (Li et al., 2026a)	Creative	×	Policy Model	2026-01	English	Hybrid	Text; Visual	Rubric Score	1,000
PresentBench (Chen et al., 2026a)	Creative	×	Policy Model	2026-03	English	Human	Text; Visual	Rubric Score	238
HelloBench (Que et al., 2024)	Creative	×	Policy Model	2024-09	English	Hybrid	Text	Rubric Score	647
WritingBench (Wu et al., 2025)	Creative	×	Policy Model	2025-03	English; Chinese	Hybrid	Text	Rubric Score	1,000
AskBench (Zhao et al., 2026)	Agentic	×	Agent Policy	2026-02	English	Model	Text	Accuracy	800
CM2 (Zhang et al., 2026f)	Agentic	✓	Agent Policy	2026-02	English	Model	Text	Rubric Reward	8,000
MoReBench (Chiu et al., 2025)	Reasoning	×	Policy Model	2025-10	English	Human	Text	Rubric Score	1,000
XIFBench (Li et al., 2025b)	Alignment	×	Policy Model	2025-03	6 languages	Hybrid	Text	Accuracy	558
InFoBench (Qin et al., 2024)	Alignment	×	Policy Model	2024-01	English	Human	Text	Accuracy	500
StrongREJECT (Souly et al., 2024)	Alignment	×	Policy Model	2024-02	English	Human	Text	Rubric Score	313
RubricBench (Zhang et al., 2026e)	Alignment	×	Rubric Generator	2026-03	English	Human	Text	F1	1,147
mR3 (Anugraha et al., 2026)	Alignment	✓	Judge Model	2025-10	72 languages	Hybrid	Text	Rubric Reward	100,000
MENLO (Whitehouse et al., 2026)	Alignment	×	Judge Model	2025-09	47 languages	Human	Text	Accuracy	6,423
MultiChallenge (Sirdeshmukh et al., 2025)	Alignment	×	Policy Model	2025-01	English	Human	Text	Accuracy	273
AdvancedIF (He et al., 2025b)	Alignment	×	Policy Model	2025-11	English	Human	Text	Accuracy	1,645
ROPD (Fang et al., 2026)	General Use	✓	Policy Model	2026-05	English	Hybrid	Text	Rubric Reward	57,000
Rubicon (Huang et al., 2025)	General Use	✓	Policy Model	2025-08	English	Hybrid	Text	Rubric Reward	5,000
III. Hybrid Evaluation Method									
FrontierScience (Wang et al., 2026)	Academic	×	Policy Model	2026-01	English	Human	Text	Rubric Score	160
PaperBench (Starace et al., 2025)	Academic	×	Agent Policy	2025-04	English	Human	Text	Rubric Score	20
PRBench-Physics (Qiu et al., 2026a)	Academic	×	Agent Policy	2026-03	English	Human	Text	Rubric Score	30
DEER (Han et al., 2025)	DeepResearch	×	Policy Model	2025-12	Korean	Human	Text	Rubric Score	50
TRAJECT-Bench (He et al., 2025a)	Agentic	×	Agent Policy	2025-10	English	Hybrid	Text	Rubric Score	5,670
RLCER (Sheng et al., 2026)	Reasoning	✓	Policy Model	2026-02	English	Model	Text	Rubric Reward	17,000

Table 7: Representative rubric-related evaluation resources grouped by evaluation method. The table reports each resource's domain, whether it is used for training, its evaluation target, release date, language coverage, rubric source, modality, evaluation indicator, and number of instances. Evaluation-method categories are shown as gray full-width group rows.

Paper	*Application Setting*	*Date*	*Task Description*	*Rubric Role*	*Eval Target*	*Rubric Source*	*Modality*	*Eval Method*
I. Medical								
Baichuan-M2 (Baichuan-M2 Team et al., 2025)	Medical	2025-09	Medical Reasoning RL	Reward Signal	Policy Model	Model	Text	LLM
InfiMed-ORBIT (Wang et al., 2025b)	Medical	2025-10	Medical Dialogue RL	Reward Signal	Policy Model	Model	Text	LLM
Health-SCORE (Yang et al., 2026c)	Medical	2026-01	Health Response RL	Reward Signal	Policy Model	Model	Text	LLM
ClinAlign (Lyu et al., 2026a)	Medical	2026-02	Clinical Alignment	Supervision Signal	Policy Model	Hybrid	Text	LLM
Quark Medical (Xu et al., 2026b)	Medical	2026-02	Medical QA RL	Reward Signal	Policy Model	Hybrid	Text	LLM
OralGPT-Plus (Fan et al., 2026a)	Medical	2026-03	Dental Tool-use RL	Reward Signal	Agent Policy	Hybrid	Text + Vision	LLM
II. Writing & Retrieval								
Well-Being Concepts (Jiang et al., 2025a)	Writing	2025-08	Well-being Eval.	Evaluation Criteria	Policy Model	Human	Text	LLM
Retro* (Lan et al., 2025)	Retrieval	2025-09	Retrieval RL	Reward Signal	Retriever	Human	Text	LLM
Pencils Down! (Farzi and Dietz, 2024)	Retrieval	2024-07	Retrieve/Generate Eval.	Evaluation Criteria	Policy Model	Hybrid	Text	LLM
LLM-Rubric (Hashemi et al., 2024)	Writing	2024-12	Text Quality Eval.	Evaluation Criteria	Policy Model	Human	Text	LLM
III. DeepResearch								
Step-DeepResearch (Hu et al., 2025)	DeepResearch	2025-12	Research Agent RL	Reward Signal	Agent Policy	Model	Text	LLM
CaRR (Zhang et al., 2026a)	DeepResearch	2026-01	Citation-aware Search RL	Reward Signal	Agent Policy	Model	Text	LLM
Query-Specific (Lv et al., 2026)	DeepResearch	2026-02	Report Rubric Gen.	Reward Signal	Rubric Generator	Hybrid	Text	LLM
DR Tulu (Shao et al., 2025)	DeepResearch	2025-11	DeepResearch RL	Reward Signal	Agent Policy	Model	Text	LLM
IV. Code								
Agentic Rubrics (Raghavendra et al., 2026)	Code	2026-01	SWE Patch Eval.	Evaluation Criteria	Agent Policy	Model	Text	LLM
StitchCUDA (Li et al., 2026c)	Code	2026-03	CUDA Code RL	Reward Signal	Agent Policy	Hybrid	Text	Hybrid
V. General Agentic								
CM2 (Zhang et al., 2026f)	Agentic	2026-02	Tool-use RL	Reward Signal	Agent Policy	Model	Text	LLM
AdaRubric (Ding, 2026)	Agentic	2026-03	Agent Trajectory RL	Reward Signal	Agent Policy	Model	Text	LLM
SYNTHAGENT (Lyu et al., 2026b)	Agentic	2026-01	Mock Tool-use RL	Reward Signal	Agent Policy	Model	Text	LLM
SCRIBE (Jiang and Ferraro, 2026)	Agentic	2026-01	Tool-use RL	Reward Signal	Agent Policy	Hybrid	Text	LLM
ARCANE (Masters et al., 2025)	Agentic	2025-12	Long-horizon Agent RL	Reward Signal	Agent Policy	Hybrid	Text	LLM
ArenaRL (Zhang et al., 2026d)	Agentic	2026-01	Open-ended Agent RL	Reward Signal	Agent Policy	Model	Text	LLM
VI. Multimodal								
RubricRL (Feng et al., 2025)	Text + Vision	2025-11	Text-to-Image RL	Reward Signal	Policy Model	Model	Text + Vision	LLM
RubiCap (Huang et al., 2026)	Text + Vision	2026-03	Captioning RL	Reward Signal	Policy Model	Model	Text + Vision	LLM
Proxy-GRM (Qiu et al., 2026b)	Text + Vision	2026-03	VLM Reward Modeling	Reward Signal	Judge Model	Model	Text + Vision	LLM
Auto-Rubric (Tian et al., 2026)	Text + Vision	2026-05	Multimodal Gen. RL	Reward Signal	Policy Model	Model	Text + Vision	LLM
rDPO (Yu et al., 2026b)	Text + Vision	2026-04	Visual Policy RL	Reward Signal	Policy Model	Model	Text + Vision	LLM
AutoRubric-R1V (Jia et al., 2025)	Text + Vision	2025-10	Multimodal Reasoning RL	Reward Signal	Policy Model	Model	Text + Vision	LLM
SpeechLLM FT (Parikh et al., 2026)	Text + Audio	2026-03	Speech Assessment FT	Supervision Signal	Policy Model	Human	Text + Audio	LLM
SpeechLLM-as-Judges (Wang et al., 2025a)	Text + Audio	2025-10	Speech Judge Eval.	Evaluation Criteria	Judge Model	Human	Text + Audio	LLM
Omni-RRM (Kong et al., 2026)	Omni-modal	2026-02	Omni Reward Modeling	Reward Signal	Judge Model	Model	Omni	LLM

Table 8: Representative rubric-based applications grouped by application setting. The table summarizes how rubrics are used across domains and modalities, including each paper's release date, task description, rubric role, evaluation target, rubric source, modality, and evaluation method. Application-setting categories are shown as gray full-width group rows.